

MYC-C7Z010-20-V2

Product Manual



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History

Version	Author	Participants	Date	Description
V1.0	Sender		2022-05-18	Initial Version
V1.1	Sender		2023-02-22	Added information about JTAG startup configuration
V1.2	Sender		2023-12-13	Change pin number and pin name

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1. Summarize

MYC-C7Z010-20-V2 development board is composed of MYC-C7Z010-20-V2 Core board and MYB-C7Z010/20 Baseboard, Xilinx Zynq-7010/7020 as the core embedded Core board. Xilinx's latest ZynQ-7000 All Programmable SoC platform based on 28nm process flow tightly integrates ARM processor and FPGA architecture. PS unit has the high performance and low power characteristics of dual-core ARM Cortex-A9 MPCore. In the design can better meet the needs of various industries. The motherboard is equipped with serial port, network port, MMC/SD/SDIO card interface, ADC interface, CAN and other interfaces, supporting Linux. The information provided includes user manual, PDF backplane schematic diagram, external expansion interface driver, BSP source code package, development tools, etc. It provides a perfect software development environment for developers, reduces the product development cycle, and realizes the rapid launch of products.

You can download the above materials at http://d.myirtech.com/MYD-C7Z010_20/

In the development stage, it is suggested to speed up the development with the evaluation kit MYC-C7Z010-20-V2 matching the Core board. Detailed information about the evaluation suite is available at: <https://www.myirtech.com/list.asp?id=698>

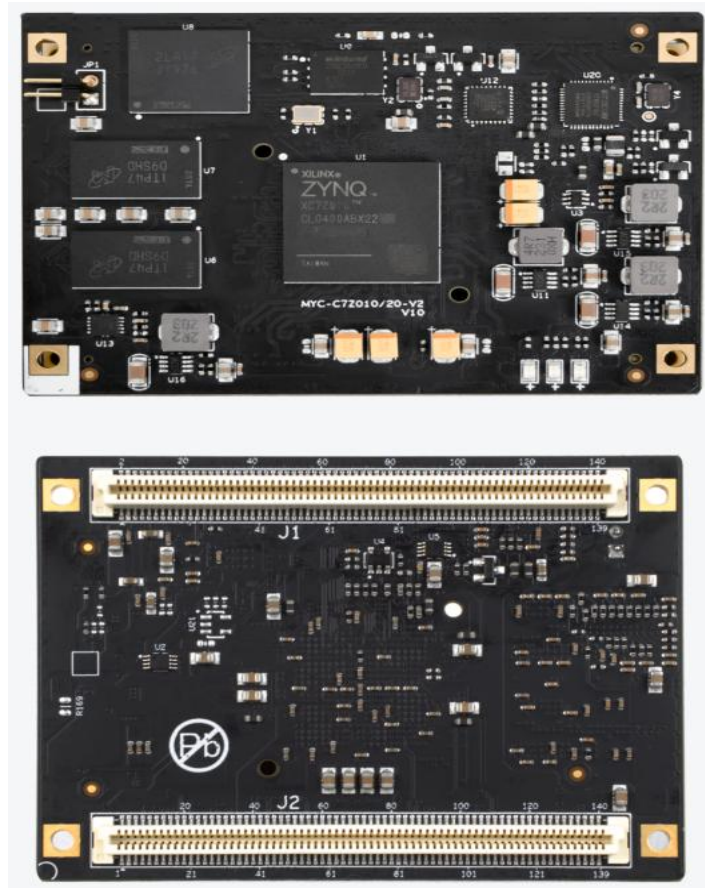


Figure 1-1 Front and back of the MYC-C7Z010-20-V2 core panel

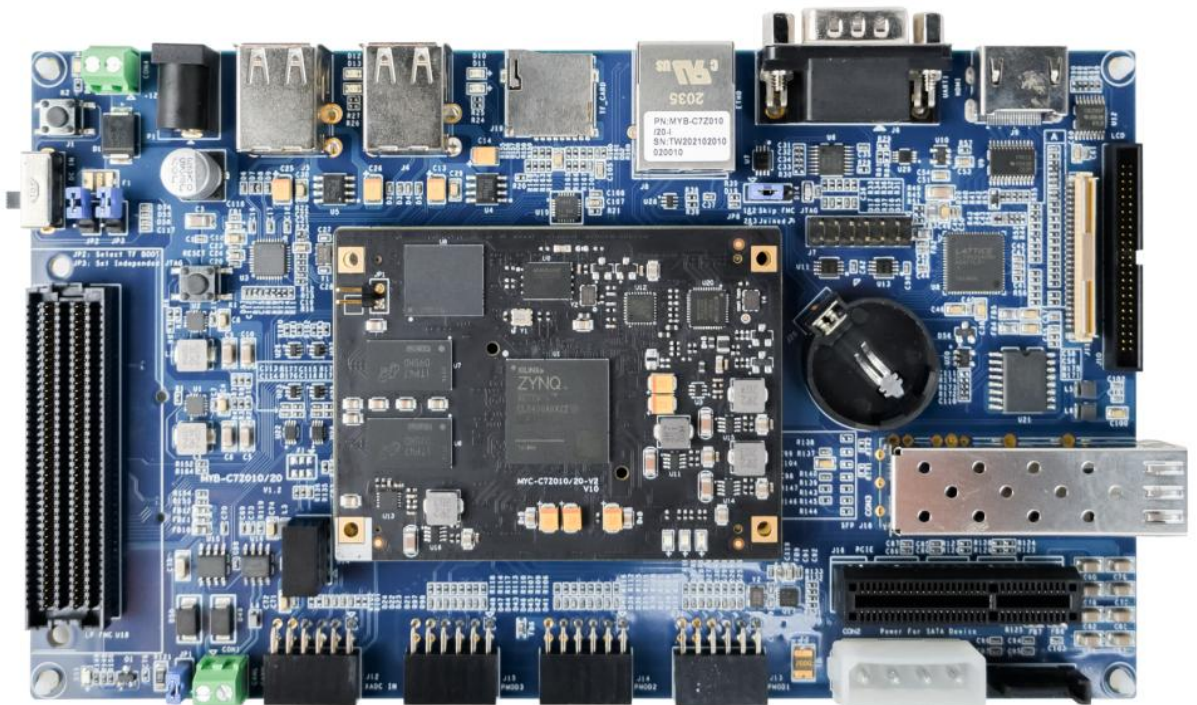


Figure 1-2 Front view of MYC-C7Z010-20-V2 evaluation kit

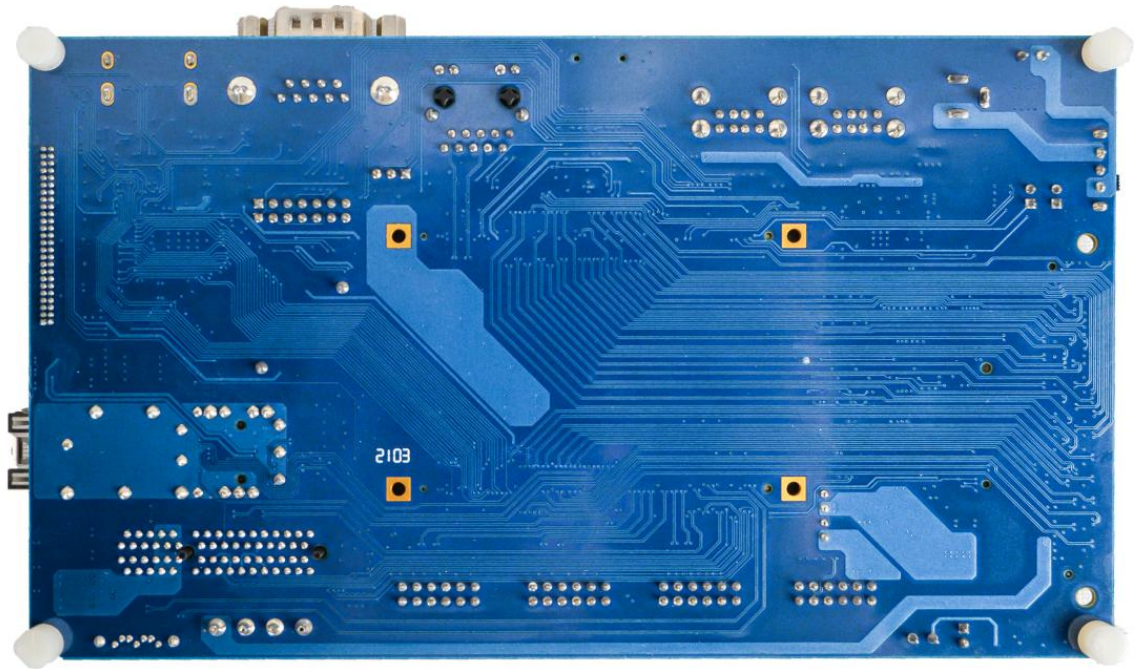


Figure 1-3 Rear of the MYC-C7Z010-20-V2 evaluation kit

2. Product presentation

MYC-C7Z010-20-V2 Core board adopts connector package, standard XC7Z010/20 processor, main frequency up to 667, 766Mhz, rich peripheral interfaces, with USB2.0, Ethernet, UART, CAN, DCMI, ADC and other common peripheral interfaces, convenient for customers to customize flexibly.

MYC-C7Z010-20-V2 series Core board contains two specific product models: they have some differences in main frequency configuration, main CPU specifications and other aspects, customers can choose the appropriate model according to their needs. For the differences between product models, see 2.4.

2.1. Chip shows

The Zynq@-7000 family is based on the Xilinx All Programmable SoC (AP SoC) architecture. These products integrate feature-rich dual-core or single-core ARM Cortex-A9 MPCore processing systems (PS) and Xilinx programmable logic (PL) on a single device, built on state-of-the-art, high performance, low power (HPL), 28 nm, high K metal gate (HKMG) process technology. The ARM Cortex-A9 MPCore processor is the core of PS, which also includes on-chip memory, external memory interfaces, and a rich set of I/O peripherals. The ZynQ-7000 family offers the flexibility and scalability of FPGAs while providing the performance, power and ease of use typically associated with ASics and ASPs. The Zynq-7000 AP SoC family of devices enables designers to use industry-standard tools on a single platform for cost-sensitive and high-performance applications. Although every device in the Zynq-7000 family contains the same PS, the PL and I/O resources are different between devices. As a result, Zynq-7000 AP SoC devices can serve a wide range of applications.

2.1.1. ZYNQ7000 Series resource comparison

ZYNQ 7000 series mainly contains processors such as Z-7010, Z-7020, and Z-7015 etc. The main resource differences among the series are shown in whether Cortex-A9 is single-core or dual-core and frequency differences. The detailed differences are shown in Table 2-1.

Feature Summary

Table 1: Zynq-7000 and Zynq-7000S SoCs

	Device Name	Z-7007S	Z-7012S	Z-7014S	Z-7010	Z-7015	Z-7020	Z-7030	Z-7035	Z-7045	Z-7100
	Part Number	XC7Z007S	XC7Z012S	XC7Z014S	XC7Z010	XC7Z015	XC7Z020	XC7Z030	XC7Z035	XC7Z045	XC7Z100
Processing System	Processor Core	Single-core ARM Cortex-A9 MPCore™ with CoreSight™			Dual-core ARM Cortex-A9 MPCore™ with CoreSight™						
	Processor Extensions	NEON™ & Single / Double Precision Floating Point for each processor									
	Maximum Frequency	667 MHz (-1); 766 MHz (-2)			667 MHz (-1); 766 MHz (-2); 866 MHz (-3)			667 MHz (-1); 800 MHz (-2); 1 GHz (-3)			667 MHz (-1) 800 MHz (-2)
	L1 Cache	32 KB Instruction, 32 KB data per processor									
	L2 Cache	512 KB									
	On-Chip Memory	256 KB									
	External Memory Support ⁽¹⁾	DDR3, DDR3L, DDR2, LPDDR2									
	External Static Memory Support ⁽¹⁾	2x Quad-SPI, NAND, NOR									
	DMA Channels	8 (4 dedicated to Programmable Logic)									
	Peripherals ⁽¹⁾	2x UART, 2x CAN 2.0B, 2x I2C, 2x SPI, 4x 32b GPIO									
	Peripherals w/ built-in DMA ⁽¹⁾	2x USB 2.0 (OTG), 2x Tri-mode Gigabit Ethernet, 2x SD/SDIO									
	Security ⁽²⁾	RSA Authentication, and AES and SHA 256-bit Decryption and Authentication for Secure Boot									
Processing System to Programmable Logic Interface Ports (Primary Interfaces & Interrupts Only)		2x AXI 32b Master 2x AXI 32-bit Slave 4x AXI 64-bit/32-bit Memory AXI 64-bit ACP 16 Interrupts									

Table 2-1 ZYNQ7000 function diagram and CPU resource differences

MYC-C7Z010-20-V2 processor main features:

ARM Cortex-A9, operating frequency 667, 766MHz
16/32 bit LPDDR-2, DDR2,DDR3L,DDR3
System level control Register (SLCR) : A set of various registers used to control PS behavior
The Snoop control unit (SCU) maintains consistency between L1 and L2
DMA controller: Four channels of PS (memory copy to/from any memory in the system)
256KB ON-chip SRAM (OCM) with parity: accessible CPU, PL and central interconnect
Single/double QUAD-SPI controller
2 PHY, 2 USB, 2 SD, 2 SPI, 2 CAN, etc
Any SMC can be the primary boot device
Accelerator Coherent Port (ACP) from PL(master) to PS(Slave)
Watchdog timer, triple counter/timer
Universal Interrupt Controller (GIC) : Allocates shared peripheral interrupt (SPI) from other parts of the system (PS and PL)
Packaging: BGA400

Table 2-2 Main features of MYC-C7Z010-20-V2

2.2. Core board Major parameter

Name	Major parameter
Main control chip series	XiLinx Zynq 7000 series
Main control chip model	XC7Z010-1CLG400 (standard configuration) XC7Z020-1CLG400 (standard configuration)
Processor specifications	667/766MHz ARM Cortex-A9
Memory	DDR3(256Mbx16) SDRAM 256Mbx16
Storage	QSPI Flash(256Mb) 、EMMC(4GB)
Core board size	75 x 55 x 8.3 mm
Interface type	Board-to-board connector, 1x2 140Pin,0.8mm
PCB specification	10 layer board design, gold sinking process
Operating system	Linux5.4.0

Table 2-3 Main parameters

2.3. System chart

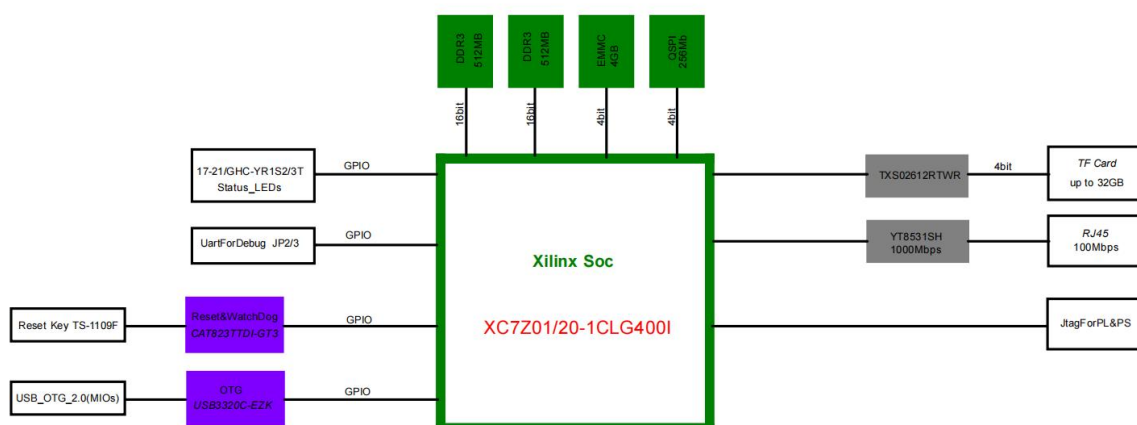


Figure 2-3 MYC-C7Z010-20-V2 Core board system block diagram

2.4. Standard model

MYC-C7Z010-20-V2 Core board is divided into two models according to CPU model, storage device, operating temperature, internal configuration and other parameters. Please select the model most suitable for you from the following list.

Model Size	MYC-C7Z010-V2-4E51D-667-I	MYC-C7Z020-V2-4E1D-766-I
Master chip	ZYNQ7000	ZYNQ7000
Main chip family	XC7Z010-1CLG400I	XC7Z020-1CLG400I
Core	Cortex-A9	Cortex-A9
Dominant frequency	667MHz	766MHz
Operating system	Linux5.4.0	Linux5.4.0
EMMC	4G	4G
Memory	1G DDR3	1G DDR3
Storage	256Mb Flash	256Mb Flash
UART	2 way	2 way
CAN	2 way	2 way
USB /OTG	2 way	2 way
Ethernet	2 way RGMII / RMII	2 way RGMII / RMII
I2C	2 way	2 way
SPI	2 way	2 way
GPIO	54 way (MAX)	54 way (MAX)
Supply voltage	+3.3V、1.8V、1.0V、1.5V	+3.3V、1.8V、1.0V、1.5V
Machine size	75*55*8.3mm	75*55*8.3mm
Operating temperature	-40°C - +85°C	-40°C - +85°C
Number of encapsulated pins	400	400
Relevant certification	CE / ROHS	CE / ROHS

Table 2-4 MYC-C7Z010-20-V2 Core board selection Table 1

Model Size	MYC-C7Z010-V2-4E1D-667-C	MYC-C7Z020-V2-4E1D-766-C
Master chip	ZYNQ7000	ZYNQ7000
Main chip family	XC7Z010-1CLG400I	XC7Z020-1CLG400I
Core	Cortex-A9	Cortex-A9
Dominant frequency	667MHz	766MHz
Operating system	Linux5.4.0	Linux5.4.0
EMMC	4G	4G
Memory	1G DDR3	1G DDR3
Storage	256Mb Flash	256Mb Flash
UART	2 way	2 way
CAN	2 way	2 way
USB /OTG	2 way	2 way
Ethernet	2 way RGMII / RMII	2 way RGMII / RMII
I2C	2 way	2 way
SPI	2 way	2 way
GPIO	54 way (MAX)	54 way (MAX)
Supply voltage	+3.3V、1.8V、1.0V、1.5V	+3.3V、1.8V、1.0V、1.5V
Machine size	75*55*8.3mm	75*55*8.3mm
Operating temperature	0°C - +70°C	0°C - +70°C
Number of encapsulated pins	400	400
Relevant certification	CE / ROHS	CE / ROHS

Table 2-4 MYC-C7Z010-20-V2 Core board selection Table 2

3. Pin description

3.1. Pin diagram

MYC-C7Z010-20-V2 Core board and Baseboard are connected with a board-to-board connector with a pitch of 0.8mm. For the packaging design of the Baseboard, refer to the instructions in Section 7.2:

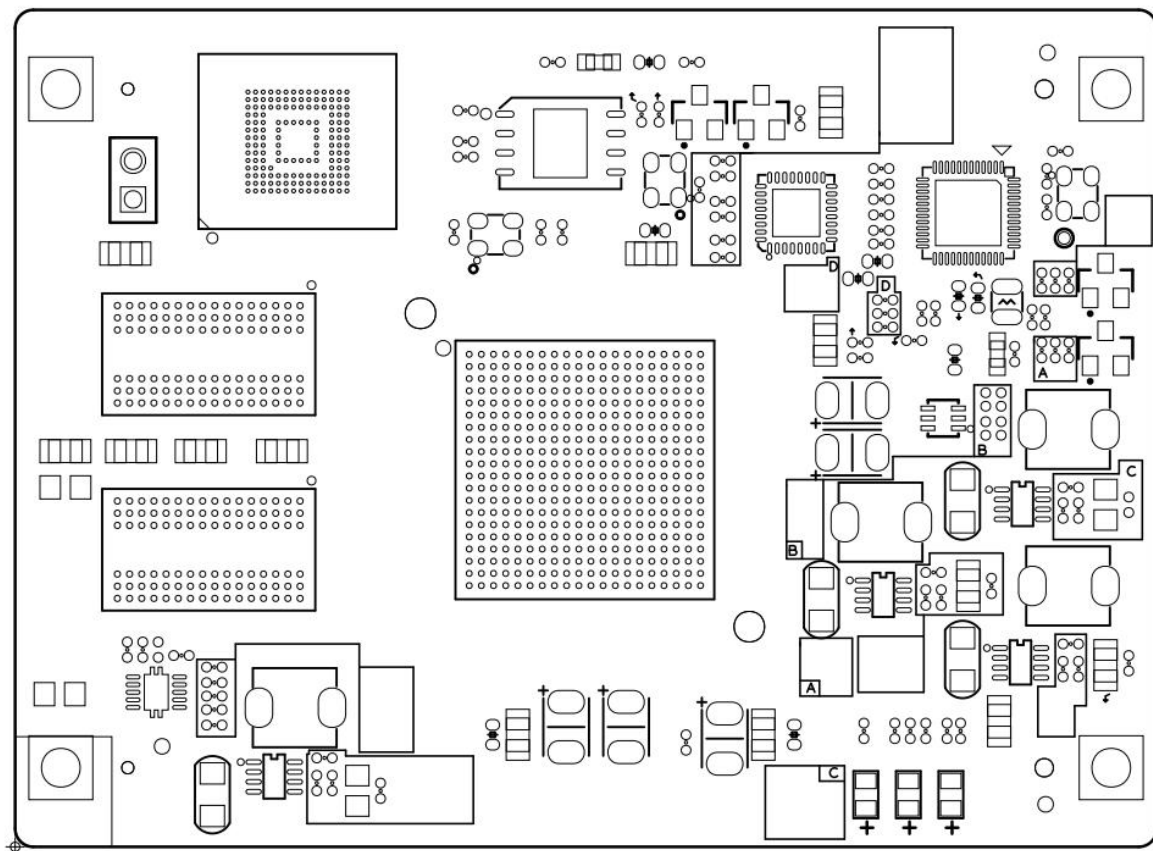


Figure 3-1 Front view of the module

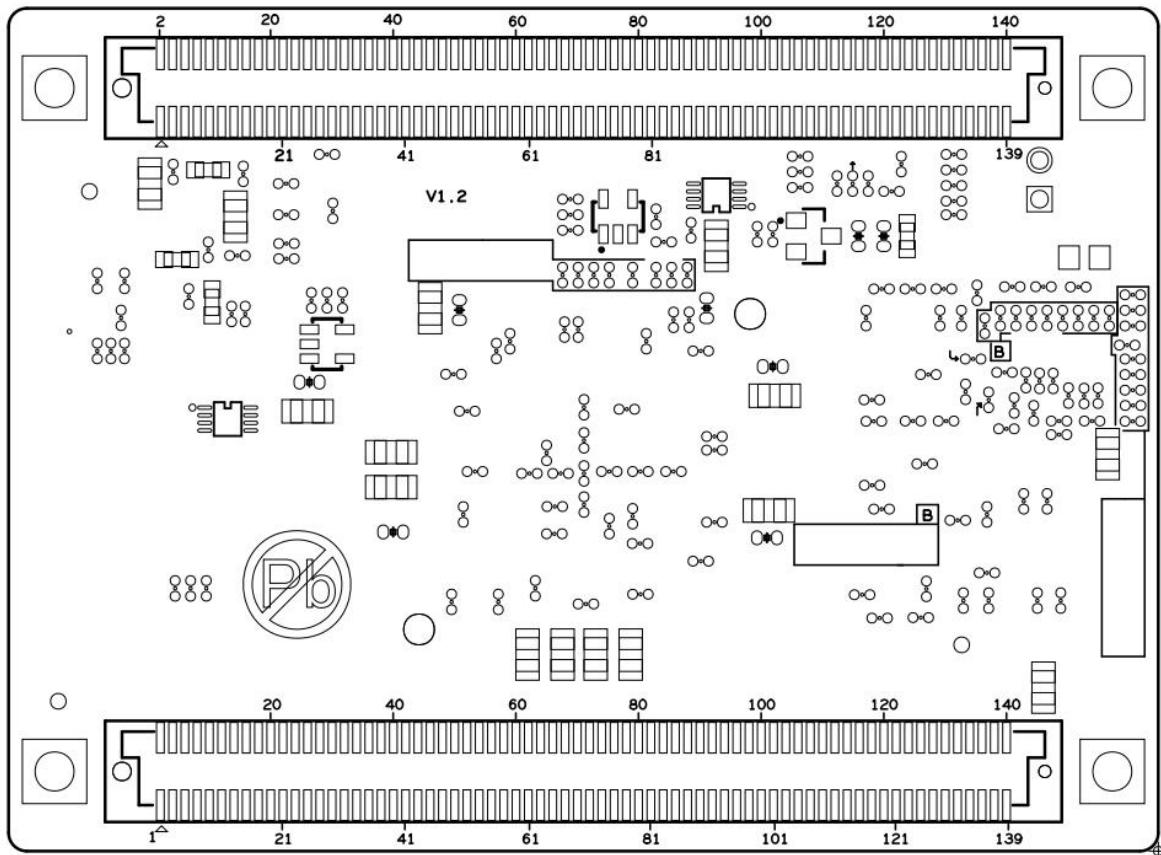


Figure 3-2 Rear view of the module

3.2. Core board pin mapping table

MYC-C7Z010-20-V2 Core board interface pins are defined in the following table. The pin functions of the BSP development package are configured according to default functions in the following table. If you need to change the default pin functions, modify the relevant driver configuration codes; otherwise, uncertain exceptions such as driver conflicts may occur.

Core board Pins	Grade	Default function	Functional description	Electrical level	Input/Output	MCU Pins	Remark
J1-P1	VDD_5V	5V Power	Power input	5V	I	—	
J1-P2	VDD_5V	5V Power	Power input	5V	I	—	
J1-P3	VDD_5V	5V Power	Power input	5V	I	—	
J1-P4	VDD_5V	5V Power	Power input	5V	I	—	
J1-P5	GND	GND	Power Ground	0V	—	—	
J1-P6	GND	GND	Power Ground	0V	—	—	
J1-P7	RGMII0_TRP0	RGMII	Differential signal	1.8V	IO	—	
J1-P8	GND	GND	Power Ground	0V	—	—	
J1-P9	RGMII0_TRN0	RGMII	Differential signal	1.8V	IO	—	
J1-P10	RGMII0_TRP1	RGMII	Differential signal	1.8V	IO	—	
J1-P11	GND	GND	Power Ground	0V	—	—	
J1-P12	RGMII0_TRN1	RGMII	Differential signal	1.8V	IO	—	
J1-P13	RGMII0_TRP2	RGMII	Differential signal	1.8V	IO	—	
J1-P14	GND	GND	Power Ground	0V	—	—	
J1-P15	RGMII0_TRN2	RGMII	Differential signal	1.8V	IO	—	
J1-P16	RGMII0_TRP3	RGMII	Differential signal	1.8V	IO	—	
J1-P17	GND	GND	Power Ground	0V	—	—	
J1-P18	RGMII0_TRN3	RGMII	Differential signal	1.8V	IO	—	
J1-P19	USB_PHY_DP	USB	Differential signal	1.8V	IO	—	
J1-P20	GND	GND	Power Ground	0V	—	—	

J1-P21	USB_PHY_DM	USB	USB Differential signal	1.8V	IO	—	
J1-P22	MDIO_CK	RGMII	RGMII MDIO_CK	1.8V	I	—	
J1-P23	GND	GND	Power Ground	0V	—	—	
J1-P24	MDIO_DATA	RGMII	MDIO_DATA	1.8V	O	—	
J1-P25	USB_PHY_ID	USB	USB_PHY_ID	1.8V	I	—	
J1-P26	RGMII0_LED_ACT	RGMII	RGMII0_LED_ACT	1.8V	O	—	
J1-P27	USB_PHY_VBUS	USB	USB_PHY_VBUS	1.8V	IO	—	
J1-P28	RGMII0_LED_LINK	RGMII	RGMII0_LED_LINK	1.8V	O	—	
J1-P29	IO_B35_0	BANK35	IO_B35_0	3.3V	IO	G14	
J1-P30	VBUS_SW_EN	USB	VBUS_SW_EN	1.8V	O	—	
J1-P31	IO_B35_LP7	BANK35	FMC_LA17_P_CC	3.3V	IO	M19	
J1-P32	GND	GND	Power Ground	0V	—	—	
J1-P33	IO_B35_LN7	BANK35	FMC_LA17_N_CC	3.3V	IO	M20	
J1-P34	IO_B35_LP18	BANK35	FMC_LA23_P	3.3V	IO	G19	
J1-P35	IO_B35_LP21	BANK35	FMC_LA13_P	3.3V	IO	N15	
J1-P36	IO_B35_LN18	BANK35	FMC_LA23_N	3.3V	IO	G20	
J1-P37	IO_B35_LN21	BANK35	FMC_LA13_N	3.3V	IO	N16	
J1-P38	IO_B35_LP9	BANK35	FMC_LA22_P	3.3V	IO	L19	
J1-P39	IO_B35_LP17	BANK35	FMC_LA14_P	3.3V	IO	J20	
J1-P40	IO_B35_LN9	BANK35	FMC_LA22_N	3.3V	IO	L20	
J1-P41	IO_B35_LN17	BANK35	FMC_LA14_N	3.3V	IO	H20	
J1-P42	IO_B35_LP11	BANK35	FMC_LA01_P_CC	3.3V	IO	L16	
J1-P43	GND	GND	Power Ground	0V	—	—	
J1-P44	IO_B35_LN11	BANK35	FMC_LA01_N_CC	3.3V	IO	L17	
J1-P45	IO_B35_LP1	BANK35	FMC_LA08_P	3.3V	IO	C20	
J1-P46	GND	GND	Power Ground	0V	—	—	
J1-P47	IO_B35_LN1	BANK35	FMC_LA08_N	3.3V	IO	B20	
J1-P48	IO_B35_LP4	BANK35	FMC_LA07_P	3.3V	IO	D19	
J1-P49	IO_B35_LP15	BANK35	FMC_LA09_P	3.3V	IO	F19	

J1-P50	IO_B35_LN4	BANK35	FMC_LA07_N	3.3V	IO	D20	
J1-P51	IO_B35_LN15	BANK35	FMC_LA07_P	3.3V	IO	F20	
J1-P52	IO_B35_LP8	BANK35	FMC_LA10_P	3.3V	IO	M17	
J1-P53	GND	GND	Power Ground	0V	—	—	
J1-P54	IO_B35_LN8	BANK35	FMC_LA10_N	3.3V	IO	M18	
J1-P55	IO_B35_LP12	BANK35	FMC_CLK0_M2C_P	3.3V	IO	K17	
J1-P56	IO_B35_LP16	BANK35	FMC_LA24_P	3.3V	IO	G17	
J1-P57	IO_B35_LN12	BANK35	FMC_CLK0_M2C_N	3.3V	IO	K18	
J1-P58	IO_B35_N16	BANK35	FMC_LA24_N	3.3V	IO	G18	
J1-P59	VDDIO_35_PL	3.3V	VDDIO_35_PL	3.3V	I	—	
J1-P60	VDDIO_35_PL	3.3V	VDDIO_35_PL	3.3V	I	—	
J1-P61	GND	GND	Power Ground	0V	—	—	
J1-P62	GND	GND	Power Ground	0V	—	—	
J1-P63	IO_B35_LP5	BANK35	FMC_LA06_P	3.3V	IO	E18	
J1-P64	IO_B35_LP10	BANK35	FMC_LA02_P	3.3V	IO	K19	
J1-P65	IO_B35_LN5	BANK35	FMC_LA06_N	3.3V	IO	E19	
J1-P66	IO_B35_LN10	BANK35	FMC_LA02_N	3.3V	IO	J19	
J1-P67	IO_B35_LP3	BANK35	FMC_LA03_P	3.3V	IO	E17	
J1-P68	IO_B35_LP14	BANK35	FMC_LA00_P_CC	3.3V	IO	J18	
J1-P69	IO_B35_LN3	BANK35	FMC_LA03_N	3.3V	IO	D18	
J1-P70	IO_B35_LN14	BANK35	FMC_LA00_N_CC	3.3V	IO	H18	
J1-P71	IO_B35_LP2	BANK35	FMC_LA04_P	3.3V	IO	B19	
J1-P72	IO_B35_LP6	BANK35	FMC_LA16_P	3.3V	IO	F16	
J1-P73	IO_B35_LN2	BANK35	FMC_LA04_N	3.3V	IO	A20	
J1-P74	IO_B35_LN6	BANK35	FMC_LA16_N	3.3V	IO	F17	
J1-P75	GND	GND	Power Ground	0V	—	—	
J1-P76	GND	GND	Power Ground	0V	—	—	
J1-P77	IO_B35_LP13	BANK35	FMC_CLK0_C2M_P	3.3V	IO	H16	
J1-P78	IO_B35_LP24	BANK35	FMC_LA15_P	3.3V	IO	K16	

J1-P79	IO_B35_LN13	BANK35	FMC_CLK0_C2M_N	3.3V	IO	H17	
J1-P80	IO_B35_LN24	BANK35	FMC_LA15_N	3.3V	IO	J16	
J1-P81	GND	GND	Power Ground	0V	—	—	
J1-P82	IO_B35_LP20	BANK35	FMC_LA20_P	3.3V	IO	K14	
J1-P83	IO_B35_LP19	BANK35	FMC_LA12_P	3.3V	IO	H15	
J1-P84	IO_B35_LN20	BANK35	FMC_LA20_N	3.3V	IO	J14	
J1-P85	IO_B35_LN19	BANK35	FMC_LA12_N	3.3V	IO	G15	
J1-P86	IO_B35_LP22	BANK35	FMC_LA19_P	3.3V	IO	L14	
J1-P87	IO_B35_LP23	BANK35	FMC_LA11_P	3.3V	IO	M14	
J1-P88	IO_B35_LN22	BANK35	FMC_LA19_N	3.3V	IO	L15	
J1-P89	IO_B35_LN23	BANK35	FMC_LA11_N	3.3V	IO	M15	
J1-P90	IO_B35_25	BANK35	FMC_SDA	3.3V	IO	J15	
J1-P91	GND	GND	Power Ground	0V	—	—	
J1-P92	GND	GND	Power Ground	0V	—	—	
J1-P93	XADC_VCC	1.8V	XADC_VCC	1.8V	O	—	
J1-P94	JTAG_TCK	BANK0	JTAG_TCK	3.3V	IO	F9	
J1-P95	XADC_INP0	BANK0	XADC_INP0	1.8V	I	K9	
J1-P96	JTAG_TMS	BANK0	JTAG_TMS	3.3V	IO	J6	
J1-P97	XADC_INN0	BANK0	XADC_INN0	1.8V	I	L10	
J1-P98	JTAG_TDI	BANK0	JTAG_TDI	3.3V	IO	G6	
J1-P99	XADC_TEMP_P	BANK0	XADC_TEMP_P	1.8V	I	M9	
J1-P100	JTAG_TDO	BANK0	JTAG_TDO	3.3V	IO	F6	
J1-P101	XADC_TEMP_N	BANK0	XADC_TEMP_N	1.8V	I	M10	
J1-P102	GND	GND	Power Ground	0V	—	—	
J1-P103	XADC_GND	XADC_GND	Power Ground	0V	—	—	
J1-P104	SDIO0_CLK	SD	SDIO0_CLK	1.8V	O	D14	
J1-P105	GND	GND	Power Ground	0V	—	—	

J1-P106	SDIO0_CMD	SD	SDIO0_CMD	1.8V	O	C17	
J1-P107	PS_MIO0	BANK500	SDIO0_D1	3.3V	O	E6	
J1-P108	SDIO0_D1	SD	S_JTAG_H	1.8V	IO	E12	
J1-P109	S_JTAG_H	BANK500	SDIO0_D2	3.3V	I	A6	
J1-P110	SDIO0_D2	SD	SD_BOOT_CONF_H	1.8V	IO	A9	
J1-P111	SD_BOOT_CONF_H	BANK500	SDIO0_D3	3.3V	I	B7	
J1-P112	SDIO0_D3	SD	PS_MIO8	1.8V	IO	F13	
J1-P113	PS_MIO8	CAN	SDIO0_D4	3.3V	O	D5	
J1-P114	SDIO0_D4	SD	PS_MIO9	1.8V	IO	B15	
J1-P115	PS_MIO9	CAN	SDIO0_CD	3.3V	O	B5	
J1-P116	SDIO0_CD	SD	PS_MIO10	1.8V	IO	D16	
J1-P117	PS_MIO10	EMMC	SDIO0_WP	3.3V	IO	E9	
J1-P118	SDIO0_WP	SD	PS_MIO11	1.8V	IO	B14	
J1-P119	PS_MIO11	EMMC	GND	3.3V	IO	C6	
J1-P120	GND	GND	PS_MIO12	0V	—	—	
J1-P121	PS_MIO12	EMMC	UART1_TX	3.3V	IO	D9	
J1-P122	UART1_TX	UART	PS_MIO13	1.8V	O	B12	
J1-P123	PS_MIO13	EMMC	UART1_RX	3.3V	IO	E8	
J1-P124	UART1_RX	UART	UART1_RX	1.8V	I	C12	
J1-P125	PS_MIO14	EMMC	PS_MIO14	3.3V	IO	C5	
J1-P126	PS_501_MIO_50	BANK500	PS_501_MIO_50	1.8V	I	B13	
J1-P127	PS_MIO15	EMMC	PS_MIO15	3.3V	IO	C8	
J1-P128	PS_501_MIO_51	BANK500	PS_501_MIO_51	1.8V	O	B9	
J1-P129	GND	GND	GND	0V	—	—	
J1-P130	FPGA_INITn	BANK0	FPGA_INITn	3.3V	O	R10	
J1-P131	Sys_PG	PG	Sys_PG	5V	0	—	
J1-P132	FPGA_WARMRESETn	BANK0	FPGA_WARMRESETn	3.3V	I	L6	
J1-P133	PS_500_RESET_OUTn	BANK500	PS_500_RESET_OUTn	1.8V	O	B7	

J1-P134	FPGA_INIT_DONE	BANK0	FPGA_INIT_DONE	3.3V	O	R11	
J1-P135	PS_WARMRESET_INn	BANK500	PS_WARMRESET_INn	1.8V	I	—	
J1-P136	VBAT_KEY_BACK_1.8V	BANK0	VBAT_KEY_BACK_1.8V	1.8V	O	F11	
J1-P137	eMMC_RSTn	EMMC	eMMC_RSTn	3.3V	I	—	
J1-P138	VDDIO_501_PS	BANK501	VDDIO_501_PS	1.8V	O	E11	
J1-P139	VDDIO_500_PS	BANK500	VDDIO_500_PS	3.3V	O	C7	
J1-P140	GND	GND	GND	0V	—	—	
J2-P1	VDD_5V	5V	VDD_5V	5V	I	—	
J2-P2	VDD_5V	5V	VDD_5V	5V	I	—	
J2-P3	VDD_5V	5V	VDD_5V	5V	I	—	
J2-P4	VDD_5V	5V	VDD_5V	5V	I	—	
J2-P5	GND	GND	GND	0V	—	—	
J2-P6	GND	GND	GND	0V	—	—	
J2-P7	IO_B34_25	BANK 34	IO_B34_25	3.3V	O	T19	
J2-P8	IO_B34_0	BANK 34	IO_B34_0	3.3V	O	R19	
J2-P9	IO_B34_LP13	BANK 34	IO_B34_LP13	3.3V	O	N18	
J2-P10	IO_B34_LP24	BANK 34	IO_B34_LP24	3.3V	O	P15	
J2-P11	IO_B34_LN13	BANK 34	IO_B34_LN13	3.3V	O	P19	
J2-P12	IO_B34_LN24	BANK 34	IO_B34_LN24	3.3V	O	P16	
J2-P13	IO_B34_LP22	BANK 34	IO_B34_LP22	3.3V	O	W18	
J2-P14	IO_B34_LP21	BANK 34	IO_B34_LP21	3.3V	O	V17	
J2-P15	IO_B34_LN22	BANK 34	IO_B34_LN22	3.3V	O	W19	
J2-P16	IO_B34_LN21	BANK 34	IO_B34_LN21	3.3V	O	V18	
J2-P17	IO_B34_LP23	BANK 34	IO_B34_LP23	3.3V	O	N17	
J2-P18	IO_B34_LP20	BANK 34	IO_B34_LP20	3.3V	O	T17	
J2-P19	IO_B34_LN23	BANK 34	IO_B34_LN23	3.3V	O	P18	
J2-P20	IO_B34_LN20	BANK 34	IO_B34_LN20	3.3V	O	R18	
J2-P21	GND	GND	Power Ground	0V	—	—	
J2-P22	GND	GND	Power Ground	0V	—	—	

J2-P23	IO_B34_LP19	BANK 34	IO_B34_LP19	3.3V	O	R16	
J2-P24	IO_B34_LP18	BANK 34	IO_B34_LP18	3.3V	O	V16	
J2-P25	IO_B34_LN19	BANK 34	IO_B34_LN19	3.3V	O	R17	
J2-P26	IO_B34_LN18	BANK 34	IO_B34_LN18	3.3V	O	W16	
J2-P27	IO_B34_LP9	BANK 34	IO_B34_LP9	3.3V	O	T16	
J2-P28	IO_B34_LP12	BANK 34	IO_B34_LP12	3.3V	O	U18	
J2-P29	IO_B34_LN9	BANK 34	IO_B34_LN9	3.3V	O	U17	
J2-P30	IO_B34_LN12	BANK 34	IO_B34_LN12	3.3V	O	U19	
J2-P31	IO_B34_LP5	BANK 34	IO_B34_LP5	3.3V	O	T14	
J2-P32	IO_B34_LP14	BANK 34	IO_B34_LP14	3.3V	IO	N20	
J2-P33	IO_B34_LN5	BANK 34	IO_B34_LN5	3.3V	IO	T15	
J2-P34	IO_B34_LN14	BANK 34	IO_B34_LN14	3.3V	O	P20	
J2-P35	VDDIO_34_PL	BANK 34	VDDIO_34_PL	3.3V	I	—	
J2-P36	VDDIO_34_PL	BANK 34	VDDIO_34_PL	3.3V	I	—	
J2-P37	GND	GND	Power Ground	0V	—	—	
J2-P38	GND	GND	Power Ground	0V	—	—	
J2-P39	IO_B34_LP11	BANK 34	FMC_LA18_P_CC	3.3V	IO	U14	
J2-P40	IO_B34_LP15	BANK 34	FMC_LA25_P	3.3V	IO	T20	
J2-P41	IO_B34_LN11	BANK 34	FMC_LA18_N_CC	3.3V	IO	U15	
J2-P42	IO_B34_LN15	BANK 34	FMC_LA25_N	3.3V	IO	U20	
J2-P43	IO_B34_LP6	BANK 34	FMC_LA32_P	3.3V	IO	P14	
J2-P44	IO_B34_LP10	BANK 34	FMC_LA26_P	3.3V	IO	V15	
J2-P45	IO_B34_LN6	BANK 34	FMC_LA32_N	3.3V	IO	R14	
J2-P46	IO_B34_LN10	BANK 34	FMC_LA26_N	3.3V	IO	W15	
J2-P47	IO_B34_LP4	BANK 34	FMC_LA33_P	3.3V	IO	V12	
J2-P48	IO_B34_LP16	BANK 34	FMC_LA24_P	3.3V	IO	V20	
J2-P49	IO_B34_LN4	BANK 34	FMC_LA33_N	3.3V	IO	W13	
J2-P50	IO_B34_LN16	BANK 34	FMC_LA24_N	3.3V	IO	W20	
J2-P51	GND	GND	Power Ground	0V	—	—	

J2-P52	GND	GND	Power Ground	0V	—	—	
J2-P53	IO_B34_LP7	BANK 34	FMC_LA30_P	3.3V	IO	Y16	
J2-P54	IO_B34_LP17	BANK 34	FMC_LA29_P	3.3V	IO	Y18	
J2-P55	IO_B34_LN7	BANK 34	FMC_LA30_N	3.3V	IO	Y17	
J2-P56	IO_B34_LN17	BANK 34	FMC_LA29_N	3.3V	IO	Y19	
J2-P57	IO_B34_LP3	BANK 34	FMC_PRSNT_M2C_L	3.3V	IO	U13	
J2-P58	IO_B34_LP8	BANK 34	FMC_LA27_P	3.3V	IO	W14	
J2-P59	IO_B34_LN3	BANK 34	HDMI_INT	3.3V	IO	V13	
J2-P60	IO_B34_LN8	BANK 34	FMC_LA27_N	3.3V	IO	Y14	
J2-P61	IO_B34_LP1	BANK 34	FMC_LA31_P	3.3V	IO	T11	
J2-P62	IO_B34_LP2	BANK 34	FMC_LA28_P	3.3V	IO	T12	
J2-P63	IO_B34_LN1	BANK 34	FMC_LA31_N	3.3V	IO	T10	
J2-P64	IO_B34_LN2	BANK 34	FMC_LA28_N	3.3V	IO	U12	
J2-P65	GND	GND	Power Ground	0V	—	—	
J2-P66	GND	GND	Power Ground	0V	—	—	
J2-P67	IO_B13_LP3	BANK 13	IO_B13_LP3	3.3V	I	—	
J2-P68	IO_B13_LP7	BANK 13	IO_B13_LP7	3.3V	IO	—	
J2-P69	IO_B13_LN3	BANK 13	IO_B13_LN3	3.3V	IO	—	
J2-P70	IO_B13_LN7	BANK 13	IO_B13_LN7	3.3V	IO	—	
J2-P71	IO_B13_LP4	BANK 13	IO_B13_LP4	3.3V	O	—	
J2-P72	IO_B13_LP8	BANK 13	IO_B13_LP8	3.3V	IO	—	
J2-P73	IO_B13_LN4	BANK 13	IO_B13_LN4	3.3V	IO	—	
J2-P74	IO_B13_LN8	BANK 13	IO_B13_LN8	3.3V	IO	—	
J2-P75	IO_B13_LP5	BANK 13	IO_B13_LP5	3.3V	I	—	
J2-P76	IO_B13_LP9	BANK 13	IO_B13_LP9	3.3V	IO	—	
J2-P77	IO_B13_LN5	BANK 13	IO_B13_LN5	3.3V	IO	—	
J2-P78	IO_B13_LN9	BANK 13	IO_B13_LN9	3.3V	IO	—	
J2-P79	GND	GND	Power Ground	0V	—	—	
J2-P80	GND	GND	Power Ground	0V	—	—	

J2-P81	VDDIO_13_PL	BANK 34	VDDIO_13_PL	3.3V	I	—	
J2-P82	VDDIO_13_PL	BANK 34	VDDIO_13_PL	3.3V	I	—	
J2-P83	IO_B13_LP21	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	V11	
J2-P84	IO_B13_0	BANK 13	IO_B13_0	3.3V	IO	V5	
J2-P85	IO_B13_LN21	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	V10	
J2-P86	IO_B13_LP16	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	W10	
J2-P87	IO_B13_LP12	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	T9	
J2-P88	IO_B13_LN16	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	W9	
J2-P89	IO_B13_LN12	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	U10	
J2-P90	IO_B13_LP20	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	Y12	
J2-P91	IO_B13_LP17	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	U9	
J2-P92	IO_B13_LN20	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	Y13	
J2-P93	IO_B13_LN17	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	U8	
J2-P94	IO_B13_LP18	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	W11	
J2-P95	GND	GND	Power Ground	0V	—	—	
J2-P96	IO_B13_LN18	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	Y11	
J2-P97	IO_B13_LP11	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	U7	
J2-P98	GND	GND	Power Ground	0V	—	—	
J2-P99	IO_B13_LN11	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	V7	
J2-P100	IO_B13_LP14	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	Y9	
J2-P101	IO_B13_LP15	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	V8	
J2-P102	IO_B13_LN14	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	Y8	
J2-P103	IO_B13_LN15	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	W8	
J2-P104	IO_B13_LP13	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	Y7	
J2-P105	IO_B13_LP22	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	V6	
J2-P106	IO_B13_LN13	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	Y6	
J2-P107	IO_B13_LN22	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	W6	
J2-P108	IO_B13_LP19	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	T5	
J2-P109	GND	GND	Power Ground	0V	—	—	

J2-P110	IO_B13_LN19	BANK 13	Only XC7Z020 Pmod IO	3.3V	IO	U5	
J2-P111	MGTPTXP0	PCIE	MGTPTXP0	3.3V	IO	—	
J2-P112	GND	GND	Power Ground	0V	—	—	
J2-P113	MGTPTXN0	PCIE	MGTPTXN0	3.3V	IO	—	
J2-P114	MGTPTXP2	PCIE	MGTPTXP2	3.3V	IO	—	
J2-P115	GND	GND	Power Ground	0V	—	—	
J2-P116	MGTPTXN2	PCIE	MGTPTXN2	3.3V	IO	—	
J2-P117	MGTPRXP0	PCIE	MGTPRXP0	3.3V	IO	—	
J2-P118	GND	GND	Power Ground	0V	—	—	
J2-P119	MGTPRXN0	PCIE	MGTPRXN0	3.3V	IO	—	
J2-P120	MGTPRXP2	PCIE	MGTPRXP2	3.3V	IO	—	
J2-P121	GND	GND	Power Ground	0V	—	—	
J2-P122	MGTPRXN2	PCIE	MGTPRXN2	3.3V	IO	—	
J2-P123	MGTPTXP1	PCIE	MGTPTXP1	3.3V	IO	—	
J2-P124	GND	GND	Power Ground	0V	—	—	
J2-P125	MGTPTXN1	PCIE	MGTPTXN1	3.3V	IO	—	
J2-P126	MGTPTXP3	PCIE	MGTPTXP3	3.3V	IO	—	
J2-P127	GND	GND	Power Ground	0V	—	—	
J2-P128	MGTPTXN3	PCIE	MGTPTXN3	3.3V	IO	—	
J2-P129	MGTPRXP1	PCIE	MGTPRXP1	3.3V	IO	—	
J2-P130	GND	GND	Power Ground	0V	—	—	
J2-P131	MGTPRXN1	PCIE	MGTPRXN1	3.3V	IO	—	
J2-P132	MGTPRXP3	PCIE	MGTPRXP3	3.3V	IO	—	
J2-P133	GND	GND	Power Ground	0V	—	—	
J2-P134	MGTPRXN3	PCIE	MGTPRXN3	3.3V	IO	—	
J2-P135	MGTREFCLKP0	PCIE	MGTREFCLKP0	3.3V	IO	—	
J2-P136	GND	GND	Power Ground	0V	—	—	
J2-P137	MGTREFCLKN0	PCIE	MGTREFCLKN0	3.3V	IO	—	
J2-P138	MGTREFCLKP1	PCIE	MGTREFCLKP1	3.3V	IO	—	

J2-P139	GND	GND	Power Ground	0V	—	—	
J2-P140	MGTREFCLKN1	PCIE	MGTREFCLKN1	3.3V	IO	—	

Table 3-1 PIN LIST of the MYC-C7Z010-20-V2 Core board

4. Electrical Specification

4.1. Main power supply

The main power supply of MYC-C7Z010-20-V2 Core board is VDD_5V, corresponding to the pins of C54 and C55 of the stamp hole. To ensure normal operation, the Baseboard must provide $5V \pm 5\%$ voltage, and ensure that the output capacity of the power supply circuit can meet the power consumption of the Core board. Section 4.3 lists the power consumption and current of the Core board under various conditions. Please reserve appropriate allowance when designing the power supply circuit. If you have higher requirements for heat dissipation, you can use a radiator.

4.2. Power domain

The external power supply voltage needs to be provided by the Baseboard, while the internal voltage is generated by the Core board itself and does not need additional power supply.

The Core board uses VDD_5V power supply, and the power management chip generates multiple different voltages to meet the power supply of MPU,DDR3,Flash and other modules. The complete power supply chip model is MP2143DJ-LF-Z.

Name	Describe	Reference voltage
VDD_5V	5V input, main supply voltage	5V

Table 4-1 External power supply voltage

4.3. Power consumption

Operating conditions	Mains input(V)	Average current(mA)	Peak current(mA)	Aotal power consumption (mW)
During boot	5	410	420	2100
Full-load	5	480	490	2400

Table 4-2 Power consumption parameters of the PSU

4.4. GPIO DC Characteristics

Parameter	Tab	Min	Recommended value	Max	Unit	Explain
High level input voltage	VIH	2.3	3.3	3.6	V	—
1.8V level input voltage	VIH	1.79V	1.8V	1.81V	V	—
Low level input voltage	VIL	-0.3	—	1.0	V	—
High level output voltage	VOH	—	3.3	—	V	—
Low level output voltage	VOL	—	—	0.9	V	—

Table 4-3 GPIO DC characteristics

5. System configuration and startup

5.1. BOOT mode setting

When the MYC-C7Z010-20-V2 series processor starts, it first executes the program in the chip's internal Boot ROM. Boot ROM starts from different Boot sources by reading Boot pins. The specific correspondence is as follows:

Boot Pin 1	Boot Pin 2	Initial Boot Source	Explain
0	0	JTAG	Boot from JTAG
0	1	QSPI	Default
1	1	Micro SD	Boot from Micro SD

Table 5-1 Processor startup mode configuration

There are two pins that determine the boot source: SD_BOOT_CONF_H and S_JTAG_H. These pins add a pull-up or pull-down design to the Core Board.

Note: When the supporting development board is used, the V12 version of JTAG mode needs to be started manually by changing the resistance R12 on the back of the core board to R10, and the jumper cap JP2 and JP3 are not closed. V10 version does not need to replace the resistor, the jumper cap is not closed.

5.2. Reset and switch

MYC-C7Z010-20-V2 core board provides two dedicated pins, PS_WARMRESET_INn and WDT_RESETN, the two functions are different, it is recommended to connect out, for different purposes.

Pin function	Explain
WDT_RESETN	Reset pin after power failure. RC reset circuit or hardware watchdog reset chip can be used to reset the output.
PS_WARMRESET_INn	And the Baseboard can be matched with key reset, to eliminate static electricity can be connected to an ESD diode.

Table 5-2 Describes the reset and Pwron pin functions

6. Interface specification

The Core board of MYC-Y7Z010-20-V2 will consider the size of PCB at the beginning of design, because the size of PCB is not enough to draw out all IO resources. A small amount of GPIO has been removed, in addition certain functions such as clock, EMMC, QSPI, Flash also take up some IO, etc. These IO are not extracted into the Core board interface. See Section 6.13 for internal resource allocation for the Core board.

It is recommended that you use Xilinx's data book and related software to allocate pin resources. For software download and instructions, visit Xilinx official website.

6.1. SDMMC

MYC-C7Z010-20-V2 Core board support leads to the MMC interface. TXS02612 chip can be used for expansion, a single SDIO port and two SDIO peripherals interface. It is recommended to use MMC1 interface for Micro SD circuits and MMC2 to use peripherals with SDIO WIFI or other SDIO interfaces.

6.1.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
D14	SDIO0_CLK	SD card	SDIO0_CLK	1.8V	O	J1-P104	
C17	SDIO0_CMD	SD card	SDIO0_CMD	1.8V	O	J1-P106	
E12	SDIO0_D1	SD card	SDIO0_D1	1.8V	IO	J1-P108	
A9	SDIO0_D2	SD card	SDIO0_D2	1.8V	IO	J1-P110	
F13	SDIO0_D3	SD card	SDIO0_D3	1.8V	IO	J1-P112	
B15	SDIO0_D4	SD card	SDIO0_D4	1.8V	IO	J1-P114	
D16	SDIO0_CD	SD card	SDIO0_CD	3.3V	IO	J1-P116	
B14	SDIO0_WP	SD card	SDIO0_WP	3.3V	IO	J1-P118	

Table 6-1 Pin definitions of MMC3 ports

6.2. UART

MYC-C7Z010-20-V2 Core board has an asynchronous serial port of up to nine pins. Due to the pin reuse of the chip, the Core board is configured with 3 pin serial port by default, in which UART1 has flow control (RTS and CTS signal) function.

6.2.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
B12	UART1_TX	UART	UART1_TX	1.8V	O	J1-P122	
C12	UART1_RX	UART	UART1_RX	1.8V	I	J1-P124	

Table 6-2 PIN definitions of UART interfaces

6.3. USB

MYC-C7Z010-20-V2 Core board provides two high-speed USB2.0 controllers, which can be configured as two HOST interfaces or one high-speed USB2.0 HOST and one high-speed USB2.0 OTG. Use USB1 port to design USB2.0 OTG. The USB transceiver is USB3320C, through which the chip is connected to the USB base Board on the Core board.

6.3.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
C18	PS_MIO39	USB1	USB1-HS Port D7	1.8V	IO		
E16	PS_MIO31	USB1	USB1-HS Port NXT	1.8V	I		
C16	PS_MIO28	USB1	USB1-HS Port D4	1.8V	IO		
C15	PS_MIO30	USB1	USB1-HS Port STP	1.8V	O		
A14	PS_MIO32	USB1	USB1-HS Port D0	1.8V	IO		
D15	PS_MIO33	USB1	USB1-HS Port D1	1.8V	IO		
C13	PS_MIO29	USB1	USB1-HS Port DIR	1.8V	I		
A12	PS_MIO34	USB1	USB1-HS Port D2	1.8V	IO		
E13	PS_MIO38	USB1	USB1-HS Port D6	1.8V	IO		

A11	PS_MIO36	USB1	USB1-HS CLK Signal	1.8V	O		
F12	PS_MIO35	USB1	USB1-HS Port D3	1.8V	IO		
A10	PS_MIO37	USB1	USB1-HS Port D5	1.8V	IO		
	USB_PHY_DP	USB	USB_PHY_DP	1.8V	IO	J1-P19	
	USB_PHY_DM	USB	USB_PHY_DM	1.8V	IO	J1-P21	
	USB_PHY_ID	USB	USB_PHY_ID	1.8V	I	J1-P25	
	USB_PHY_VBUS	USB	USB_PHY_VBUS	1.8V	I	J1-P27	
	VBUS_SW_EN	USB	VBUS_SW_EN	1.8V	O	J1-P30	

Table 6-3 Definitions of USB port pins

6.4. Ethernet

MYC-C7Z010-20-V2 Core board supports two gigabit network ports. Users need to design PHY chip circuit and network port socket to achieve network communication. Gigabit Ethernet Controller (GEM) implements 10/100/1000MB /s Ethernet MAC, is compliant with IEEE 802.3-2008 standards, and is capable of running in half-duplex or full-duplex mode at all three speeds. The controller provides an MDIO interface for PHY management. PHYS can be controlled from either of the MDIO interfaces.

6.4.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electric level	Input/Output	Core board pins	Remark
C11	PS_MIO53_501	PHY	LINK_MDIO_DATA	1.8V	O		
A17	PS_MIO20_501	PHY	LINK_RGMII0_TXD3	1.8V	I		
D10	PS_MIO19_501	PHY	LINK_RGMII0_TXD2	1.8V	I		
B18	PS_MIO18_501	PHY	LINK_RGMII0_TXD1	1.8V	I		
E14	PS_MIO17_501	PHY	LINK_RGMII0_TXD0	1.8V	I		
A19	PS_MIO16_501	PHY	LINK_RGMII0_TX_CLK	1.8V	I		
F14	PS_MIO21_501	PHY	LINK_RGMII0_TX_EN	1.8V	I		

A15	PS_MIO26_501	PHY	LINK_RGMII0_RXD3	1.8V	O		
B17	PS_MIO22_501	PHY	LINK_RGMII0_RX_CLK	1.8V	O		
C7	PS_POR_B_500	PHY	PS_PWRON_RESETn	1.8V	O		
F15	PS_MIO25_501	PHY	LINK_RGMII0_RXD2	1.8V	O		
A16	PS_MIO24_501	PHY	LINK_RGMII0_RXD1	1.8V	O		
D11	PS_MIO23_501	PHY	LINK_RGMII0_RXD0	1.8V	O		
D13	PS_MIO27_501	PHY	LINK_RGMII0_RX_EN	1.8V	O		
C10	PS_MIO52_501	PHY	LINK_MDIO_CLK	1.8V	O		
	RGMII0_TRP0	PHY	RGMII0_TRP0	1.8V	O	J1-P7	
	RGMII0_TRN0	PHY	RGMII0_TRN0	1.8V	O	J1-P9	
	RGMII0_TRP1	PHY	RGMII0_TRP1	1.8V	O	J1-P10	
	RGMII0_TRN1	PHY	RGMII0_TRN1	1.8V	O	J1-P12	
	RGMII0_TRP2	PHY	RGMII0_TRP2	1.8V	O	J1-P13	
	RGMII0_TRN2	PHY	RGMII0_TRN2	1.8V	O	J1-P15	
	RGMII0_TRP3	PHY	RGMII0_TRP3	1.8V	O	J1-P16	
	RGMII0_TRN3	PHY	RGMII0_TRN3	1.8V	O	J1-P18	

Table 6-4 Ethernet pin definitions

6.5. QSPI

MYC-C7Z010-20-V2 Core board supports a maximum of two QSPI controllers in master/slave mode. SPI signals include SPI_CLK, SPI_MOSI and SPI_MISO. During design, the relationship between master and slave devices should be confirmed first, and then the direction of MOSI and MISO signals should be confirmed. One QSPI interface is configured on the core board by default. If you want to use more QSPI interfaces, refer to the chip manual for configuration.

6.5.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
B8	QSPI0_D0	QSPI	QSPI0_D0	3.3V	IO		
D6	QSPI0_D1	QSPI	QSPI0_D1	3.3V	IO		
A7	QSPI0_CS0	QSPI	QSPI0_CS0	3.3V	IO		
A5	QSPI0_SCK	QSPI	QSPI0_SCK	3.3V	IO		

Table 6-5 PIN definitions of QSPI ports

6.6. JTAG

The standard JTAG interface consists of four lines :TMS, TCK, TDI and TDO, which are mode selection, clock, data input and data output lines respectively. PL-JTAG Interface clock PL-JTAG uses the JTAG port to connect to the XADC interface and uses the JTAG clock TCK. JTAG boot mode is considered a slave boot mode and is always an insecure boot mode. The JTAG chain can be configured in cascading and independent modes..

6.6.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
F9	JTAG_TCK	JTAG	JTAG clock signal	3.3V	IO	J1-P94	
G6	JTAG_TDI	JTAG	JTAG input signal	3.3V	IO	J1-P96	
F6	JTAG_TDO	JTAG	JTAG output signal	3.3V	IO	J1-P98	
J6	JTAG_TMS	JTAG	JTAG chip select signal	3.3V	IO	J1-P100	

Table 6-6 JTAG pin definitions

6.7. HDMI

MYC-C7Z010-20-V2 design HDMI interface, high definition multimedia interface is a digital video/audio interface technology, is suitable for image transmission of the special type of digital interface, it can simultaneously transmit audio and image signals, the highest data transmission speed of 2.25GB/s, No d/D or D/D conversion is required prior to signal transmission.

6.7.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
T14	IO_B34_LP5	HDMI	12 bit input pixel data bus	3.3V	I	J2-P31	
T15	IO_B34_LN5	HDMI	12 bit input pixel data bus	3.3V	I	J2-P33	
T16	IO_B34_LP9	HDMI	12 bit input pixel data bus	3.3V	I	J2-P27	
U17	IO_B34_LN9	HDMI	12 bit input pixel data bus	3.3V	I	J2-P29	
U18	IO_B34_LP12	HDMI	12 bit input pixel data bus	3.3V	I	J2-P28	
U19	IO_B34_LN12	HDMI	12 bit input pixel data bus	3.3V	I	J2-P30	
V16	IO_B34_LP13	HDMI	12 bit input pixel data bus	3.3V	I	J2-P9	
W16	IO_B34_LN13	HDMI	12 bit input pixel data bus	3.3V	I	J2-P11	
V16	IO_B34_LP18	HDMI	12 bit input pixel data bus	3.3V	I	J2-P24	
W16	IO_B34_LN18	HDMI	12 bit input pixel data bus	3.3V	I	J2-P26	
R16	IO_B34_LP19	HDMI	12 bit input pixel data bus	3.3V	I	J2-P23	
R17	IO_B34_LN19	HDMI	12 bit input pixel data bus	3.3V	I	J2-P25	
T17	IO_B34_LP20	HDMI	12 bit input pixel data bus	3.3V	I	J2-P18	
R18	IO_B34_LN20	HDMI	12 bit input pixel data bus	3.3V	I	J2-P20	
V17	IO_B34_LP21	HDMI	12 bit input pixel data bus	3.3V	I	J2-P14	
V18	IO_B34_LN21	HDMI	12 bit input pixel data bus	3.3V	I	J2-P16	
W18	IO_B34_LP22	HDMI	12 bit input pixel data bus	3.3V	I	J2-P13	
W19	IO_B34_LN22	HDMI	12 bit input pixel data bus	3.3V	I	J2-P15	
N17	IO_B34_LP23	HDMI	12 bit input pixel data bus	3.3V	I	J2-P17	
P18	IO_B34_LN23	HDMI	12 bit input pixel data bus	3.3V	I	J2-P19	
V13	IO_B34_LN3	HDMI	Interrupt output	3.3V	0	J2-P52	
D8	PS_MIO7	HDMI	reset signal	3.3V	I	J1-P133	

Table 6-7 HDMI pin definitions

6.8. LCD

MYC-C7Z010-20-V2 designs a channel of 2x25_2.54mm spacing double row pin interface, which defines multiple GPIO ports,LCD signals,FPGA signals, 3.3V,DC_IN_5V on this pin row. The interface figure is as follows.

6.8.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electric level	Input/Output	Core board pins	Remark
T14	IO_B34_LP5	HDMI	Display data signal	3.3V	IO	J2-P31	
T15	IO_B34_LN5	HDMI	Display data signal	3.3V	IO	J2-P33	
T16	IO_B34_LP9	HDMI	Display data signal	3.3V	IO	J2-P27	
U17	IO_B34_LN9	HDMI	Display data signal	3.3V	IO	J2-P29	
U18	IO_B34_LP12	HDMI	Display data signal	3.3V	IO	J2-P28	
U19	IO_B34_LN12	HDMI	Display data signal	3.3V	IO	J2-P30	
V16	IO_B34_LP13	HDMI	Enable display data	3.3V	IO	J2-P9	
W16	IO_B34_LN13	HDMI	Display clock signal	3.3V	IO	J2-P11	
V16	IO_B34_LP18	HDMI	Display data signal	3.3V	IO	J2-P24	
W16	IO_B34_LN18	HDMI	Display data signal	3.3V	IO	J2-P26	
R16	IO_B34_LP19	HDMI	Display data signal	3.3V	IO	J2-P23	
R17	IO_B34_LN19	HDMI	Display data signal	3.3V	IO	J2-P25	
T17	IO_B34_LP20	HDMI	Display data signal	3.3V	IO	J2-P18	
R18	IO_B34_LN20	HDMI	Display data signal	3.3V	IO	J2-P20	
V17	IO_B34_LP21	HDMI	Display field signal	3.3V	IO	J2-P14	
V18	IO_B34_LN21	HDMI	Display line signal	3.3V	IO	J2-P16	
W18	IO_B34_LP22	HDMI	Display data signal	3.3V	IO	J2-P13	
W19	IO_B34_LN22	HDMI	Display data signal	3.3V	IO	J2-P15	
N17	IO_B34_LP23	HDMI	Display data signal	3.3V	IO	J2-P17	
P18	IO_B34_LN23	HDMI	Display data signal	3.3V	IO	J2-P19	
P15	IO_B34_LP24	HDMI	Capacitive touch reset	3.3V	O	J2-P10	
P16	IO_B34_LN24	HDMI	Capacitive touch interrupt	3.3V	O	J2-P12	
T19	IO_B34_25	HDMI	Active high level	3.3V	IO	J2-P7	
R19	IO_B34_0	HDMI	backlight power PWM control	3.3V	O	J2-P8	

Table 6-8 LCD pin definitions

6.9. FMC

MYC-C7Z010-20-V2 is designed with the FMC interface, which provides standard sandwich board (sub card) sizes, connectors and module interfaces for FPGAs on the base board (carrier card). In this way, the I/O interface is separated from FPGA, which not only simplifies the I/O interface module design, but also maximizes the reuse of the load card. This model is ASP-134603-01. The specifications are Array Socket SKT 160Pin 1.27mm Solder. It has compatibility: standardized power supply, standardized signal definition, increase compatibility with each other.

6.9.1. Pin definition

Note: because there are too many pins, the description is not expanded here, and several special descriptions are listed. The detailed pin information can be viewed in its specification and schematic diagram.

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
F9	JTAG_TCK	FMC	JTAG_TCK	3.3V	IO		
J6	JTAG_TMS	FMC	JTAG_TMS	3.3V	IO		
G6	JTAG_TDI	FMC	JTAG_TDI	3.3V	I		
F6	JTAG_TDO	FMC	JTAG_TDO	3.3V	O		
	JTAG_TRST	FMC	FMC_JTAG_TRST	3.3V	IO		
T11	IO_B34_LP1	FMC	FMC differential signals	3.3V	IO	J2-P61	
T10	IO_B34_LN1	FMC	FMC differential signals	3.3V	IO	J2-P63	
T12	IO_B34_LP2	FMC	FMC differential signals	3.3V	IO	J2-P62	
U12	IO_B34_LN2	FMC	FMC differential signals	3.3V	IO	J2-P64	
U13	IO_B34_LP3	FMC	FMC_PRSENT	3.3V	IO	J2-P57	
C20	IO_B35_LP1	FMC	FMC differential signals	3.3V	IO	J1-P45	
B20	IO_B35_LN1	FMC	FMC differential signals	3.3V	IO	J1-P47	
B19	IO_B35_LP2	FMC	FMC differential signals	3.3V	IO	J1-P71	
A20	IO_B35_LN2	FMC	FMC differential signals	3.3V	IO	J1-P73	
E17	IO_B35_LP2	FMC	FMC differential signals	3.3V	IO	J1-P67	
D18	IO_B35_LN2	FMC	FMC differential signals	3.3V	IO	J1-P69	

Table 6-9 PIN definitions of FMC ports

6.10. FPC

MYC-C7Z010-20-V2 is designed with one FPC connector. Connector model fh12-50s-0.5 SV, 50Pin,0.5mm, 3.9mm Height. FPC has the characteristics of small size, light weight and thin thickness.

6.10.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
T14	IO_B34_LP5	HDMI	Display data signal	3.3V	IO		
T15	IO_B34_LN5	HDMI	Display data signal	3.3V	IO		
T16	IO_B34_LP9	HDMI	Display data signal	3.3V	IO		
U17	IO_B34_LN9	HDMI	Display data signal	3.3V	IO		
U18	IO_B34_LP12	HDMI	Display data signal	3.3V	IO		
U19	IO_B34_LN12	HDMI	Display data signal	3.3V	IO		
V16	IO_B34_LP13	HDMI	Enable display data	3.3V	IO		
W16	IO_B34_LN13	HDMI	Display clock signal	3.3V	IO		
V16	IO_B34_LP18	HDMI	Display data signal	3.3V	IO		
W16	IO_B34_LN18	HDMI	Display data signal	3.3V	IO		
R16	IO_B34_LP19	HDMI	Display data signal	3.3V	IO		
R17	IO_B34_LN19	HDMI	Display data signal	3.3V	IO		
T17	IO_B34_LP20	HDMI	Display data signal	3.3V	IO		
R18	IO_B34_LN20	HDMI	Display data signal	3.3V	IO		
V17	IO_B34_LP21	HDMI	Display field signal	3.3V	IO		
V18	IO_B34_LN21	HDMI	Display line signal	3.3V	IO		
W18	IO_B34_LP22	HDMI	Display data signal	3.3V	IO		
W19	IO_B34_LN22	HDMI	Display data signal	3.3V	IO		
N17	IO_B34_LP23	HDMI	Display data signal	3.3V	IO		
P18	IO_B34_LN23	HDMI	Display data signal	3.3V	IO		
P15	IO_B34_LP24	HDMI	Capacitive touch reset	3.3V	O		
P16	IO_B34_LN24	HDMI	touch interrupt	3.3V	O		
T19	IO_B34_25	HDMI	Active high level	3.3V	IO		
R19	IO_B34_0	HDMI	backlight power PWM control	3.3V	O		

Table 6-10 PIN definitions of FPC interfaces

6.11. RTC battery socket port

MYC-C7Z010-20-V2 has a button battery holder. The battery seat is 2Pinx1.25mm spacing. Button batteries with rated output voltage of 3.0V are required. Control in I2C mode.

6.11.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
B13	PS_501_MIO_50	CAN	RTC_I2C_SCL	3.3V	I	J1-P126	
B9	PS_501_MIO_51	CAN	RTC_I2C_SDA	3.3V	O	J1-P128	

Table 6-11 RTC clock pin definitions

6.12. CAN

MYC-C7Z010-20-V2 supports two CAN controllers. Comply with ISO 11898-1, CAN 2.0A, CAN2.0B standards. Supports standard frames (11-bit identification) and extended frames (29-bit identification). Supports 1mb /s bit rate.

6.12.1. Pin definition

MCU Pins	Grade	Default function	Functional description	Electrical level	Input/Output	Core board pins	Remark
D5	PS_MIO8	CAN	CAN_TXD	3.3V	I	J1-P113	
B5	PS_MIO9	CAN	CAN_RXD	3.3V	O	J1-P115	

Table 6-12 PIN definitions of the CAN interface

6.13. GPIO

The Universal I/O (GPIO) peripheral provides software to observe and control up to 54 device pins through the MIO module. It also provides access to 64 inputs from programmable logic (PL) and 128 outputs from the EMIO interface to PL. GPIO is organized into four sets of registers that group related interface signals. Each GPIO is independently and dynamically programmed as input, output, or interrupt sensing. Software can read all GPIO values within a bank using a single load instruction, or write data to one or more GPIO(within the scope of GPIO) using a single storage instruction. GPIO controls and state registers are memory-mapped at base address 0xE000 A000.

6.13.1. Pin definition

GPIO module is divided into four groups:

Bank0: 32-bit Bank control MIO pin [31:0]

Bank1: 22-bit Bank control MIO pin [53:32]

Note :Bank1 is limited to 22 bits because MIO has a total of 54 pins

Bank2: 32-bit Bank control EMIO signal [31:0]

Bank3: 32-bit Bank control EMIO signal [63:32]

The software controls GPIO through a series of memory-mapped registers. The controls for each Bank are the same, although there are some minor differences between MIO and EMIO banks due to their different functions.

6.14. Core board internal resource pin description

MYC-C7Z010-20-V2 There are some IO resources inside the Core board, these pins have been set to a bit fixed function, can not change the resource allocation of these pins.

In addition, considering that users may want to define Core board resources more freely when developing products based on The MYIR Core board, please exclude the following pins in defining resources, and do not define other pins in the same way as the following default functions.

MPU Pin labels	Default function	Remark
E9	eMMC_D0	EMMC
E8	eMMC_D1	
C5	eMMC_D2	
C8	eMMC_D3	
C6	eMMC_CMD	
D9	eMMC_CLK	
U13	IO_B34_LP3	LED indicator light
R11	FPGA_CONFIG_DONE	LED indicator light D7
E6	WDT_FEED	watchdog
D8	PS_500_RESET_OUTn	restoration
A19	RGMII_TX_CLK	Network Chip
E14	RGMII_TXD0	Network Chip
B18	RGMII_TXD1	Network Chip
D10	RGMII_TXD2	Network Chip
A17	RGMII_TXD3	Network Chip
F14	RGMII_TX_EN	Network Chip
B17	RGMII_RX_CLK	Network Chip
D11	RGMII_RXD0	Network Chip
A16	RGMII_RXD1	Network Chip
F15	RGMII_RXD2	Network Chip
A15	RGMII_RXD3	Network Chip
D13	RGMII_RX_EN	Network Chip

C10	MDIO_CLK	Network Chip
C11	MDIO_DATA	Network Chip
MIO09		unused
MIO50		unused
MIO51		unused

Table 6-14 Definitions of internal resource pins of the Core board

7. Packaging information

7.1. Machine dimension

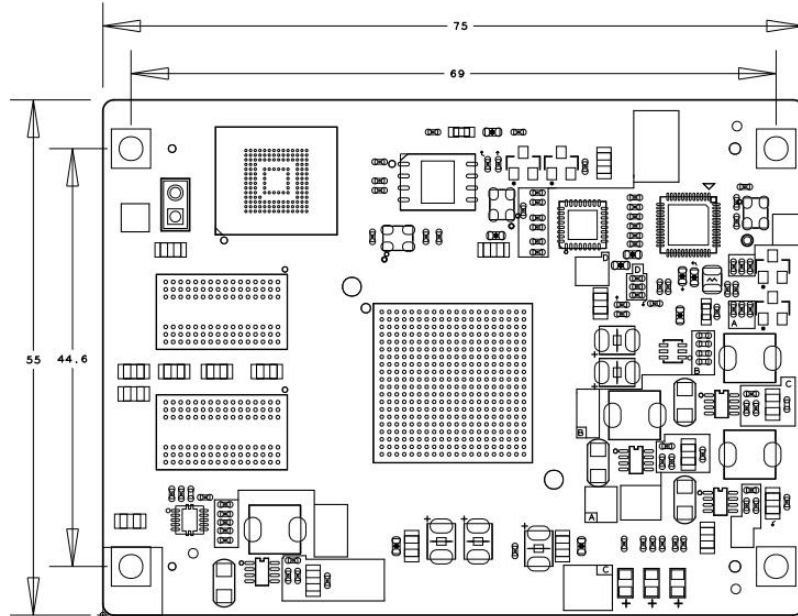


Figure 7-1 Top view of the MYC-C7Z010-20-V2 Core board (unit: mm)

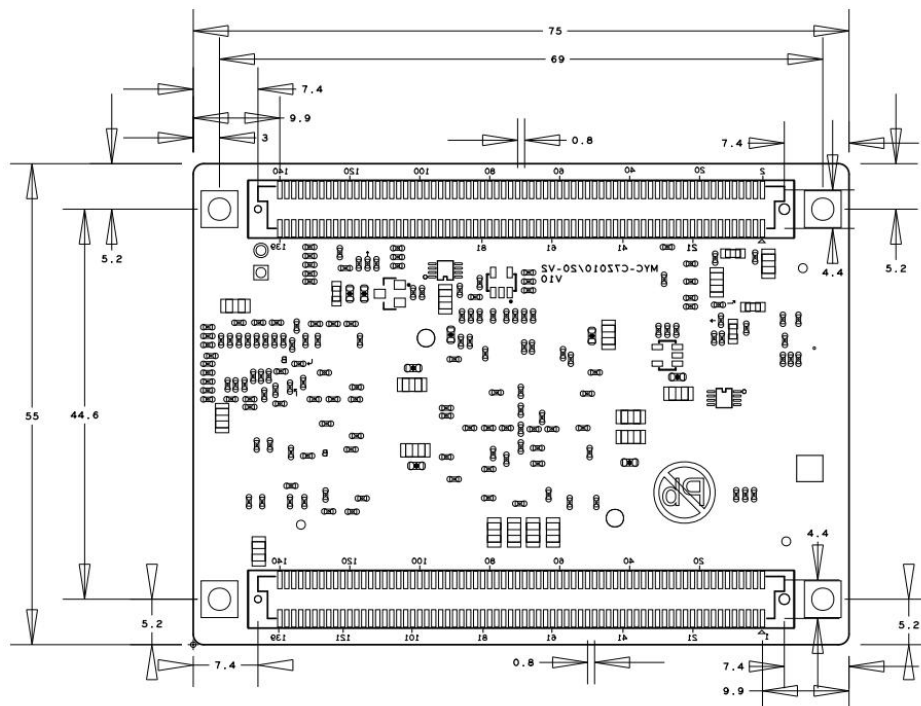


Figure 7-2 Bottom view of the MYC-C7Z010-20-V2 Core board (unit: mm)

7.2. Backboard PCB package

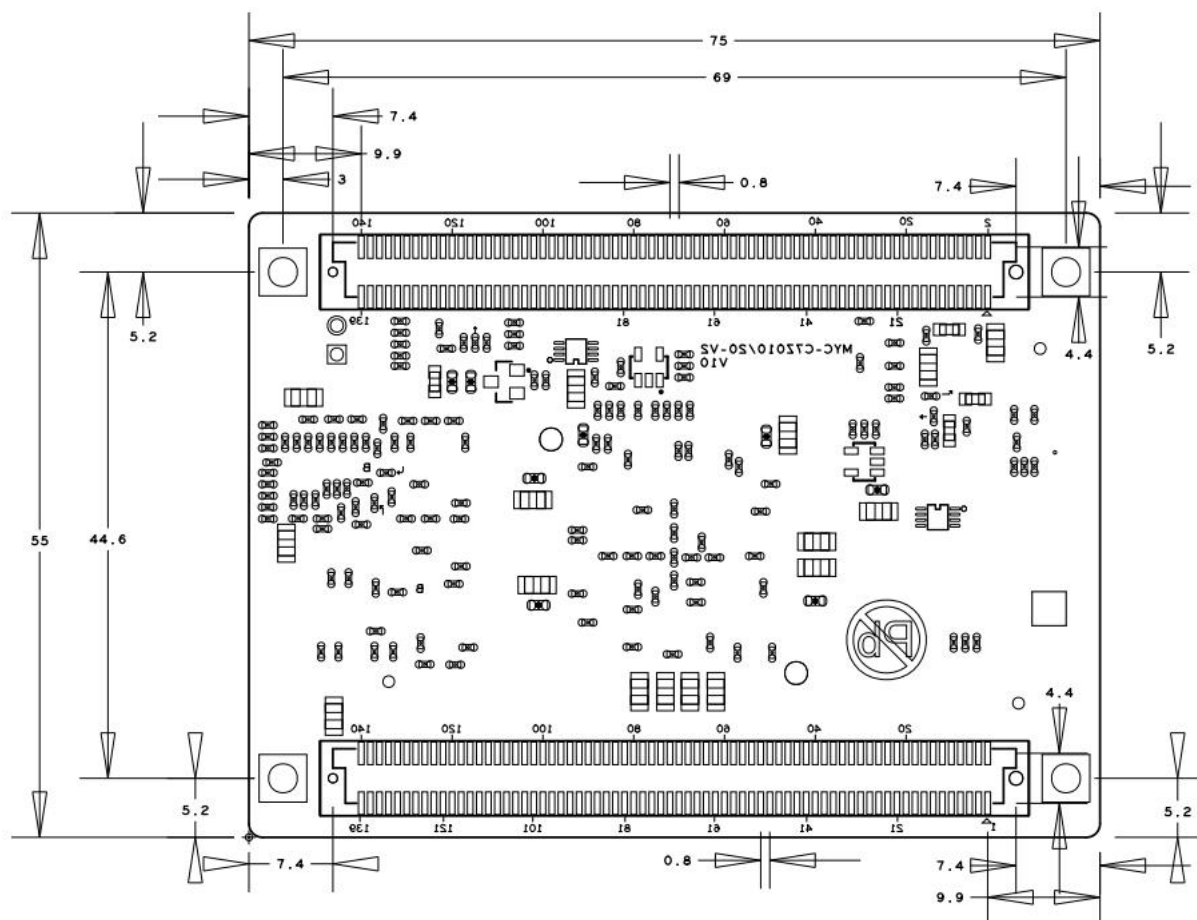


Figure 7-3 PCB package of MYC-C7Z010-20-V2 Core board (unit: mm)

Connector pin 140 Pin pitch 0.8mm。 MYIR Electronics provides designed PCB packages. Please visit http://d.myirtech.com/MYD-C7Z010_20/ to obtain the package file.

7.3. PCB Requirements

A) The thickness of PCB is recommended to be at least 1.6mm, and the balance of copper coating should be paid attention to. If PCB deformation occurs after passing through the furnace, it is recommended to use a carrier to fix passing through the furnace.

B) To ensure the quality of mounting and tin, please ensure that the distance between modules and other components on the PCB is at least 3mm.

C) Please design the package of the Core board module according to Section 7.2, or use the PCB package provided by MYIR Electronics.

Appendix A

Warranty & Technical Support Services

MYIR Electronics Limited is a global provider of ARM hardware and software tools, design solutions for embedded applications. We support our customers in a wide range of services to accelerate your time to market.

MYIR is an ARM Connected Community Member and work closely with ARM and many semiconductor vendors. We sell products ranging from board level products such as development boards, single board computers and CPU modules to help with your evaluation, prototype, and system integration or creating your own applications. Our products are used widely in industrial control, medical devices, consumer electronic, telecommunication systems, Human Machine Interface (HMI) and more other embedded applications. MYIR has an experienced team and provides custom design services based on ARM processors to help customers make your idea a reality.

The contents below introduce to customers the warranty and technical support services provided by MYIR as well as the matters needing attention in using MYIR' s products.

Service Guarantee

MYIR regards the product quality as the life of an enterprise. We strictly check and control the Core board design, the procurement of components, production control, product testing, packaging, shipping and other aspects and strive to provide products with best quality to

customers. We believe that only quality products and excellent services can ensure the long-term cooperation and mutual benefit.

Price

MYIR insists on providing customers with the most valuable products. We do not pursue excess profits which we think only for short-time cooperation. Instead, we hope to establish long-term cooperation and win-win business with customers. So we will offer reasonable prices in the hope of making the business greater with the customers together hand in hand.

Delivery Time

MYIR will always keep a certain stock for its regular products. If your order quantity is less than the amount of inventory, the delivery time would be within three days; if your order quantity is greater than the number of inventory, the delivery time would be always four to six weeks. If for any urgent delivery, we can negotiate with customer and try to supply the goods in advance.

Technical Support

MYIR has a professional technical support team. Customer can contact us by email (support@myirtech.com), we will try to reply you within 48 hours. For mass production and customized products, we will specify person to follow the case and ensure the smooth production.

After-sale Service

MYIR offers one year free technical support and after-sales maintenance service from the purchase date. The service covers:

Technical support service

MYIR offers technical support for the hardware and software materials which have provided to customers;

- To help customers compile and run the source code we offer;
- To help customers solve problems occurred during operations if users follow the user manual documents;
- To judge whether the failure exists;
- To provide free software upgrading service.

- However, the following situations are not included in the scope of our free technical support service:
- Hardware or software problems occurred during customers' own development;
- Problems occurred when customers compile or run the OS which is tailored by themselves;
- Problems occurred during customers' own applications development;
- Problems occurred during the modification of MYIR's software source code.

After-sales maintenance service

The products except LCD, which are not used properly, will take the twelve months free maintenance service since the purchase date. But following situations are not included in the scope of our free maintenance service:

- The warranty period is expired;
- The customer cannot provide proof-of-purchase or the product has no serial number;
- The customer has not followed the instruction of the manual which has caused the damage the product;
- Due to the natural disasters (unexpected matters), or natural attrition of the components, or unexpected matters leads the defects of appearance/function;
- Due to the power supply, bump, leaking of the roof, pets, moist, impurities into the boards, all those reasons which have caused the damage of the products or defects of appearance;
- Due to unauthorized weld or dismantle parts or repair the products which has caused the damage of the products or defects of appearance;
- Due to unauthorized installation of the software, system or incorrect configuration or computer virus which has caused the damage of products.

Warm tips:

- MYIR does not supply maintenance service to LCD. We suggest the customer first check the LCD when receiving the goods. In case the LCD cannot run or no display, customer should contact MYIR within 7 business days from the moment get the goods.
- Please do not use finger nails or hard sharp object to touch the surface of the LCD.

- MYIR suggests user purchasing a piece of special wiper to wipe the LCD after long time use, please avoid clean the surface with fingers or hands to leave fingerprint.
- Do not clean the surface of the screen with chemicals.
- Please read through the product user manual before you using MYIR' s products.
- For any maintenance service, customers should communicate with MYIR to confirm the issue first. MYIR' s support team will judge the failure to see if the goods need to be returned for repair service, we will issue you RMA number for return maintenance service after confirmation.

Maintenance period and charges

MYIR will test the products within three days after receipt of the returned goods and inform customer the testing result. Then we will arrange shipment within one week for the repaired goods to the customer. For any special failure, we will negotiate with customers to confirm the maintenance period.

For products within warranty period and caused by quality problem, MYIR offers free maintenance service; for products within warranty period but out of free maintenance service scope, MYIR provides maintenance service but shall charge some basic material cost; for products out of warranty period, MYIR provides maintenance service but shall charge some basic material cost and handling fee.

Shipping cost

During the warranty period, the shipping cost which delivered to MYIR should be responsible by user; MYIR will pay for the return shipping cost to users when the product is repaired. If the warranty period is expired, all the shipping cost will be responsible by users.

Products Life Cycle

MYIR will always select mainstream chips for our design, thus to ensure at least ten years continuous supply; if meeting some main chip stopping production, we will inform customers in time and assist customers with products updating and upgrading.

Value-added Services

- MYIR provides services of driver development base on MYIR' s products, like serial port, USB, Ethernet, LCD, etc.
- MYIR provides the services of OS porting, BSP drivers ' development, API software development, etc.
- MYIR provides other products supporting services like power adapter, LCD panel, etc.
- ODM/OEM services.

MYIR Electronics Limited

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